

LABREPORT

[METAL1: DEFORMATION AND STRENGTHENING]



NAME (ID NO.)	Che Xuanming
LAB PARTNERS	Ma Yige; Wang Gaoke; Liu Jiuru; Liu Chunyang; Yao Yuanhao
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LAB EXPERIMENT	Bubble Raft Experiments

Part I – Bubble raft

Introduction

Metallic crystals are among the earliest studied and most widely used classes of solid materials. From the late 19th to the early 20th century, techniques such as X-ray diffraction were used to reveal the periodic, ordered arrangement of atoms inside metals, and Bragg's law, proposed by *W. H.* and *W. L. Bragg* provided a theoretical foundation for crystal structure characterisation. Subsequent research has shown that macroscopic mechanical properties such as yield behaviour, strength and ductility are closely related to the internal atomic arrangement and the presence of crystal defects.

In the ideal case, a metallic crystal can be regarded as a perfect structure in which atoms occupy lattice points in a strictly regular manner. However, real materials inevitably contain various types of **crystal defects**, including point defects (such as vacancies and substitutional or interstitial solute atoms), line defects (such as dislocations) and planar defects (such as grain boundaries and phase boundaries). These defects play a decisive role in **plastic deformation**: the generation and glide of dislocations are the primary mechanisms by which metals deform plastically, while interactions between dislocations and their pinning at obstacles such as solute atoms, precipitates and grain boundaries collectively determine the strength, work hardening behaviour and ductility of metals.

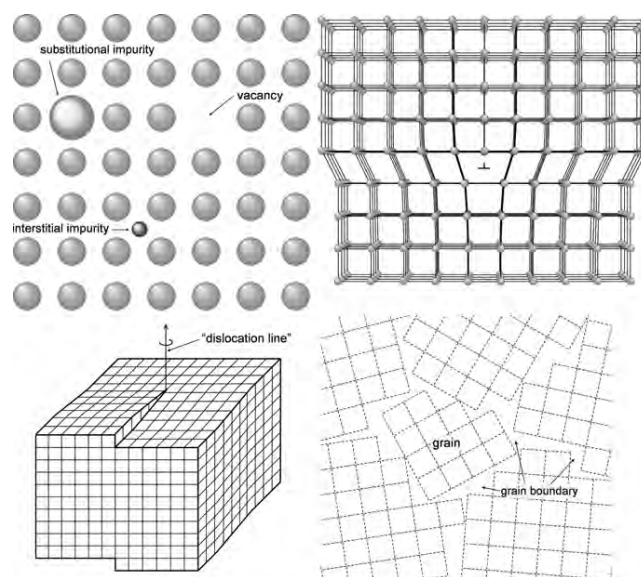


Figure 1. Crystal structures showing point defects (substitutional or interstitial elements, vacancies), line defects (edge and screw dislocations), planar defects (grain boundaries)

Because of the extremely small atomic length scale, it is very difficult to directly observe the structure and motion of defects in real metals. To address this, a variety of analogue models have been proposed, such as using small magnets or discs floating on a water surface to represent “atoms”, which can self-assemble into crystal-like structures under mutual interactions. However, these models typically suffer from large friction, a limited number of available “atoms”, and difficulty in constructing large, ordered arrays. To overcome these limitations, *Bragg* and his colleagues proposed **the bubble raft model** in the 1940s: by preparing millimetre-sized soap bubbles in a surfactant solution and allowing them to self-assemble under surface tension and bubble–bubble interactions, one can obtain arrays that spontaneously form hexagonal close-packed structures analogous to the {111} planes of face-centred cubic metals. These bubbles are numerous, experience negligible mutual friction and have relatively long lifetimes, making them well suited as a macroscopic “visualized crystal”.

The aim of the present experiment is to use the bubble raft model to simulate metallic crystal structures and their typical defect behaviour. A soap solution composed of water, dishwashing liquid and glycerol is prepared, and a single-layer bubble raft is generated in a shallow tray using a capillary–pump system to obtain “crystalline” regions with long-range hexagonal order. On this basis, by adjusting the preparation conditions and applying controlled disturbances to the raft, we observe and record the formation and evolution of defects such as vacancies, solute-like atoms (size-mismatched bubbles), grain boundaries and dislocations. Using the acquired images and videos, we qualitatively analyse the local strain fields introduced by different defects and their interactions with dislocation motion, thereby deepening our understanding of plastic deformation in metals and of strengthening mechanisms such as solid-solution strengthening and grain-refinement strengthening.

Materials and Equipment



Figure 2. Materials and Equipment in this experiment.

- (1) **Water:** Used as the solvent for preparing the soap solution.
- (2) **Dishwashing soap:** Acts as a surfactant to generate stable bubbles; a type with minimal additives should be selected.
- (3) **Glycerin:** Used to extend bubble lifetime and reduce evaporation.
- (4) **Beaker:** Used for mixing and holding the soap solution.
- (5) **Glass rod:** Used to stir the liquids to make the solution uniform.
- (6) **Fish tank air pump and plastic tube:** Provide a stable airflow to generate bubbles.
- (7) **Capillary tube (inner diameter ≈ 0.1 mm):** Connected to the end of the plastic tube and used to form small, uniform bubbles.
- (8) **Hot melt glue gun:** Used to seal the connection between the capillary tube and the plastic tube to prevent air leakage.
- (9) **Shallow pan:** Used to hold the soap solution and to form a single-layer bubble raft.
- (10) **Smartphone:** Used to record the formation, arrangement and motion of the bubbles, and to observe the evolution of dislocations and other defects.

Procedures

- (1) Take a clean beaker and add water, dishwashing liquid and glycerin in a volume ratio of 10:1:1. Stir gently until the mixture is completely homogeneous, avoiding vigorous stirring that would generate bubbles, until a uniform, transparent soap solution is obtained.
- (2) Insert the glass capillary tube into the end of the plastic tube and seal the joint uniformly with hot melt glue. After the glue has solidified, check for air leakage to ensure that the connection is secure and airtight, then connect the other end of the plastic tube to the outlet of the air pump.
- (3) Slowly pour the prepared soap solution into the shallow pan, adjusting the liquid layer thickness to about 1 cm. Use a spoon to gently skim off large bubbles on the surface so that the liquid surface becomes smooth and essentially bubble-free.
- (4) Switch on the air pump and gently immerse the tip of the capillary tube into the soap solution. Adjust the airflow and slowly move the capillary position so that bubbles are generated uniformly over the liquid surface, until a single-layer bubble raft with a reasonably uniform arrangement is formed.
- (5) Once a stable bubble structure has been established, observe and identify vacancies (missing bubbles), “solute atoms” (individual bubbles with clearly different sizes) and grain boundaries (interfaces between regions of different lattice orientation). Use a smartphone to photograph the bubble arrangement; then gently disturb the edge of the bubble raft and record videos capturing dislocation glide and its interactions with other defects.
- (6) At the end of the experiment, turn off the air pump, remove the capillary tube and rinse it thoroughly. Discard the remaining soap solution, clean the shallow pan and beaker, and tidy up all experimental equipment.

Results and observations

(1) Vacancy

Figure 3. shows a nearly perfect hexagonal close-packed array of bubbles, within which an isolated missing bubble can be clearly identified. This “hole” does not remain as an empty gap: the six neighbouring bubbles relax inwards towards the centre, so that the local lattice spacing is reduced and the originally straight lattice rows become slightly curved around the defect. Compared with the ideal hexagonal schematic in **Figure 3(a)**, the experimental bubble raft in **Figure 3(b)** therefore provides a direct visualisation of a vacancy-type point defect. The inward displacement of the surrounding bubbles indicates that removal of a single “atom” breaks the local force balance and generates a small, but clearly observable, strain field in the lattice. This observation from our own photographs demonstrates how vacancies distort an otherwise ordered array, analogous to atomic vacancies in crystalline materials.

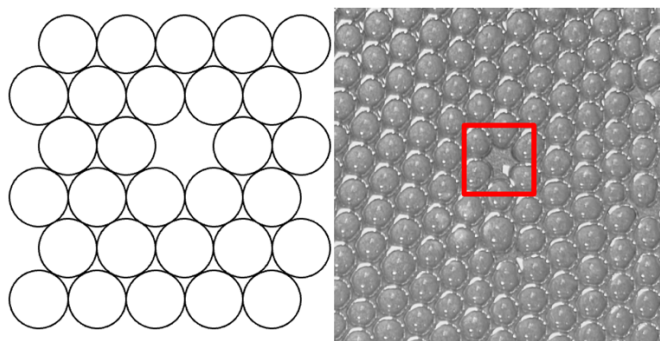


Figure 3. Vacancy in a bubble raft: (a) schematic hexagonal lattice showing a missing “atom”; (b) experimental bubble raft image, where inward relaxation of neighbouring bubbles around the vacancy is visible.

(2) Solute Atoms

A. Large Solute Atoms

Figure 4. shows a single bubble with a markedly larger diameter than its neighbours within an otherwise regular hexagonal array. As highlighted in **Figure 4(a)**, this “oversized” bubble locally breaks the geometric packing condition and forces the surrounding bubbles to move radially outwards. The corresponding experimental image in **Figure 4(b)** shows an expanded local lattice spacing and a slight bending of

the lattice rows around the defect, consistent with the formation of a compressive strain field centred on the large bubble. This configuration is analogous to a large substitutional solute atom in a crystalline lattice: the size mismatch between solute and solvent atoms introduces local compressive stress, which perturbs the otherwise uniform lattice and impedes dislocation motion. Our observation therefore provides a direct visual model for the size-misfit mechanism underlying solid solution strengthening in metallic materials.

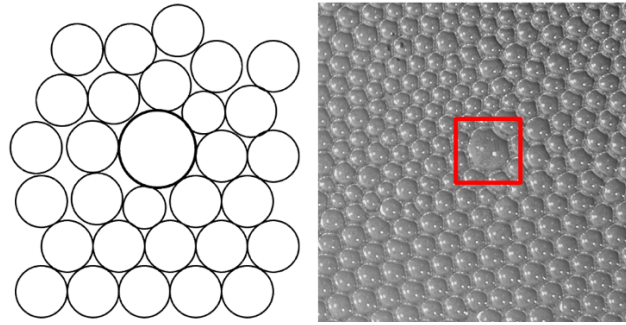


Figure 4. Large solute bubble: (a) schematic illustration of a large substitutional “solute atom” in a hexagonal lattice; (b) experimental bubble raft image showing an oversized bubble producing a local compressive distortion of the surrounding lattice.

B. Small Solute Atoms

Figure 5. shows, a much smaller bubble—essentially an incompletely filled void—was observed embedded within the otherwise uniform hexagonal packing. As indicated in **Figure 5(a)**, the neighbouring bubbles contract inwards in order to maintain contact with the undersized bubble, leading to a local reduction in bubble–bubble spacing and a subtle curvature of the surrounding lattice rows. The experimental image in **Figure 5(b)** therefore visualises a tensile-type strain field centred on the small “solute” bubble, analogous to the lattice distortion introduced by undersized substitutional or interstitial atoms in a crystalline solid. Such tensile strain fields interact elastically with moving dislocations and tend to pin or slow their motion, providing a simple physical model of solid solution strengthening arising from small solute atoms.

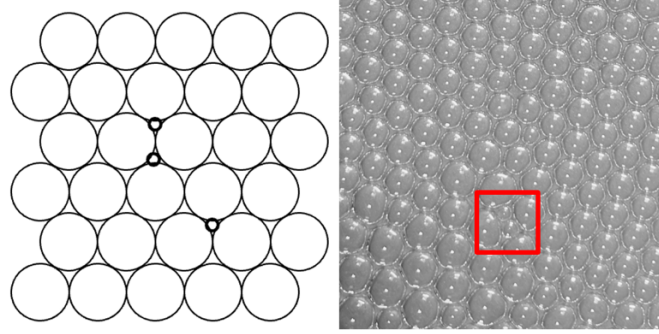


Figure 5. Small solute bubble: (a) schematic illustration of a small substitutional or interstitial “solute atom”; (b) experimental bubble raft image showing a smaller bubble inducing a local tensile-type distortion in the surrounding lattice.

(3) Grain Boundary

In the bubble raft, two crystalline regions with clearly different lattice orientations were obtained, as shown in **Figure 6**. At the interface between these domains, the regular hexagonal packing is disrupted and the bubbles lose long-range order, forming a structurally disordered grain boundary. This planar defect is directly visible as a narrow band in which rows of bubbles are kinked and misaligned relative to the adjoining grains. In analogy with crystalline materials, such grain boundaries impede the glide of dislocations, so that dislocations tend to pile up at these interfaces and the overall strength of the material is increased via grain refinement strengthening. Under sufficiently high applied stress, however, grain boundaries can also act as sources of new dislocations, which may be emitted into the neighbouring grains. Our observations therefore highlight the dual role of grain boundaries—both as barriers to and sources of dislocations—arising from their relatively high energy and structural instability.

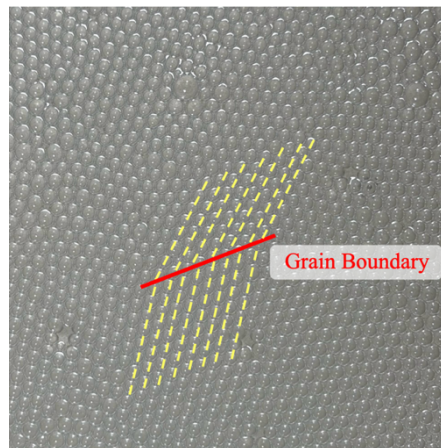


Figure 6. Grain boundary in the bubble raft.

(4) Movement of Dislocations (Core Observation)

Figure 7. shows the glide of an edge dislocation was clearly captured as a sequential, wave-like propagation of the terminating half-plane of bubbles under an applied shear disturbance. The dislocation line, identified by the abrupt termination of several close-packed rows, advanced stepwise across the lattice rather than through rigid sliding of entire bubble layers. When the moving dislocation approached regions containing “solute” bubbles of mismatched size or intersected a grain boundary, its motion was noticeably slowed or temporarily arrested, indicating strong interactions with the associated strain fields and planar defects. These real-time recordings provide a direct visual analogue of plastic deformation in crystalline metals, in which macroscopic permanent shape change is accommodated by dislocation glide. The observed hindrance of dislocation motion by solute-induced strain fields and grain boundaries illustrates the microscopic basis of solid solution strengthening and grain refinement strengthening: greater external stress is required to overcome these barriers, which manifests macroscopically as an increase in material strength.

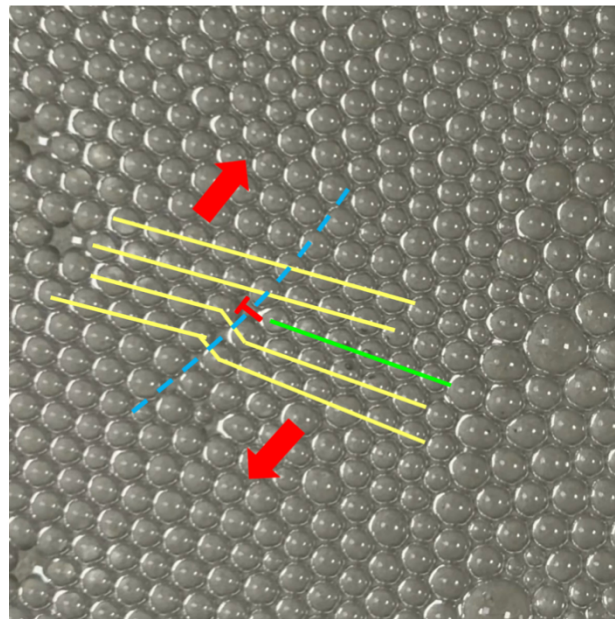


Figure 7. The movement of dislocation in the bubble raft

Your contributions

In this group experiment, I played a central role in both the practical work and the subsequent analysis. During the laboratory session, I was responsible for assembling the bubble raft apparatus, including sealing the capillary tube to the plastic tubing with hot melt glue and checking for air leakage to ensure a stable and controllable bubble flow. I also took the lead in generating a uniform single-layer bubble raft and in systematically recording high-quality photographs and videos of the bubbles under different conditions. In particular, I identified and documented key defect types such as vacancies, size-mismatched “solute” bubbles, grain boundaries and edge dislocations, and captured clear footage of dislocation glide under gentle perturbation. After the experiment, I worked closely with my teammates to interpret these observations, relate them to the underlying mechanisms of plastic deformation and strengthening, and contributed substantially to the drafting and refinement of the Results and Discussion sections of the report.

Conclusions

This experiment successfully produced a high-quality bubble raft with well-ordered hexagonal regions, enabling clear visualisation of key crystal defects, including vacancies, size-mismatched “solute” bubbles, grain boundaries and, most importantly, the glide motion of edge dislocations. The observed local lattice distortions around point defects, the structurally disordered interfacial regions at grain boundaries and the stepwise propagation of dislocations under shear were all consistent with theoretical descriptions of point, line and planar defects in crystalline materials.

By analysing how moving dislocations interacted with solute-type bubbles and grain boundaries, we directly confirmed the physical basis of solid-solution strengthening and grain-refinement strengthening: defects that generate local strain fields or act as planar barriers were observed to impede dislocation glide, implying that higher applied stresses are required to continue plastic deformation. Overall, the experiment achieved its intended purpose of transforming abstract concepts of crystal defects and strengthening mechanisms into tangible, visual phenomena, thereby deepening our understanding of the relationship between microstructural features and the mechanical behaviour of metals.